



Title: *Hydrocarbon-Degrading Microbial Consortia for Oil Drilling Waste Treatments in Arid Environments*

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Abstract:

Bioremediation is a promising, environmentally friendly approach for treating oil drilling waste, a challenge that is particularly acute in arid regions with intensive petroleum production. Effective bioaugmentation strategies require microbial consortia that are naturally adapted to such harsh environmental conditions. In this study, four metal(loid)-resistant, hydrocarbon-degrading microbial consortia capable of growing at 40 °C were isolated from oil drilling waste subjected to different phytomanagement conditions. These consortia displayed distinct microbial compositions and were able to degrade between 15% and 35% of total petroleum hydrocarbons within 15 days. The degradation processes produced varying hydrocarbon fraction profiles, supported by the identification of 14 specific OTUs determined through linear discriminant analysis effect size (LEfSe). Each consortium was characterized by groups of genera correlated with particular hydrocarbon fractions, accounting for their differences in degradation efficiency and resulting fraction profiles. Consequently, these consortia could be applied individually or sequentially to develop effective bioremediation strategies for managing multi-contaminated oil drilling waste in arid environments.

Biography

Samia Haddadi is a researcher with a PhD from the University of Pau and Pays de l'Adour, working in an Algerian oil company. Her work focuses on microbial diversity in arid ecosystems impacted by petroleum pollution. She characterizes microbial communities associated with endemic desert plants and other species thriving under extreme aridity. These stress-adapted microbial assemblages contribute to contaminant transformation, ecological stability, and resilience. By identifying climate-resistant microbial diversity, she develops bioprocesses that restore degraded arid environments under growing climatic and industrial pressures..

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